

Heterotrimeric G-protein shuttling via Gip1 extends the dynamic range of eukaryotic chemotaxis

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Chemotactic eukaryotic cells can sense chemical gradients over a wide range of concentrations via heterotrimeric G-protein signaling; however, the underlying wide-range sensing mechanisms are only partially understood. Here we report that a novel regulator of G proteins, G protein-interacting protein 1 (Gip1), is essential for extending the chemotactic range of *Dictyostelium* cells. Genetic disruption of Gip1 caused severe defects in gradient sensing and directed cell migration at high but not low concentrations of chemoattractant. Also, Gip1 was found to bind and sequester G proteins in cytosolic pools. Receptor activation induced G-protein translocation to the plasma membrane from the cytosol in a Gip1-dependent manner, causing a biased redistribution of G protein on the membrane along a chemoattractant gradient. These findings suggest that Gip1 regulates G-protein shuttling between the cytosol and the membrane to ensure the availability and biased redistribution of G protein on the membrane for receptor-mediated chemotactic signaling. This mechanism offers an explanation for the wide-range sensing seen in eukaryotic chemotaxis.

eukaryotic chemotaxis | gradient sensing | dynamic range extension | heterotrimeric G protein

Chemotaxis in eukaryotic cells is observed in many physiological processes including embryogenesis, neuronal wiring, wound healing, and immune responses (1, 2). Chemotactic cells share basic properties including high sensitivity to shallow gradients and responsiveness to a wide dynamic range of chemoattractants (3, 4). For instance, human neutrophils and *Dictyostelium* cells can sense spatial differences in chemoattractant concentration across the cell body in shallow gradients as low as 2% and exhibit chemotaxis over a 10^5 – 10^6 -fold range of background concentrations (5–7). Thus, wide-range sensing and adaptation are critical features of chemotaxis as well as other sensory systems such as visual signal transduction (8). However, the underlying regulatory mechanisms in eukaryotic chemotaxis remain unclear.

The molecular mechanisms of chemotaxis are evolutionarily conserved among many eukaryotes that use G protein-coupled receptors (GPCRs) and heterotrimeric G proteins to detect chemoattractant gradients (3, 4). In *Dictyostelium* cells, extracellular cAMP works as a chemoattractant, and binding to its receptor cyclic AMP receptor 1 (cAR1) activates G proteins ($G\alpha 2G\beta\gamma$) along the concentration gradient, leading to the activation of multiple signaling cascades including the PI3K–PTEN, TorC2–PDK–PKB, phospholipase A2, and guanylyl cyclase pathways. In contrast to the spatial distributions of cAMP/cAR1 association and G-protein activation, downstream signaling pathways are activated in an extremely biased manner at the anterior or posterior of the cell (3, 4). For example, localized patches of phosphatidylinositol 3,4,5-trisphosphate (PIP_3) are generated at the plasma membrane by an intracellular signal transduction excitable network (STEN) and function as a cue to control the pseudopod formation of motile cells (9, 10). Because PIP_3 patches have a relatively constant size of a few microns in diameter, this excitable mechanism can ensure a constant output of chemotactic responses over a wide range of concentrations. However, it is unclear how chemical gradients are sensed

adaptively over a wide range in the signal transduction cascades upstream of STEN.

Insight into this question is provided by bacterial chemotaxis and other sensory systems, such as photoreceptor rhodopsin (8). Chemoreceptor methylation in bacteria confers a broad chemotactic range (11). In light adaptation, the phosphorylation of rhodopsins in the visual system leads to rhodopsin down-regulation by arrestin, which blocks physical interaction with G-protein transducin (12). Phosphorylation-dependent receptor internalization is a feature of other systems for suppressing intracellular responses (13). Overall, in these sensory systems, the chemical modifications of receptors are important for regulating the dynamic range of the response. Consistently, *Dictyostelium* cells expressing unphosphorylated mutant cAR1 exhibit a narrow chemotactic range (14), and phosphorylated cAR1s have reduced affinity for cAMP (15). Thus, chemical modifications of chemoattractant receptors are also important in eukaryotic chemotaxis as a mechanism to extend the chemotactic range. In addition to the receptor modifications, G proteins are phosphorylated and recruited from the cytosol to the plasma membrane upon receptor stimulation in *Dictyostelium* cells (16, 17), although the relevance of these actions on wide-range sensing and adaptation is unknown.

Here we report that a novel regulator of G proteins, G protein-interacting protein 1 (Gip1), is essential for the wide-range chemotaxis in *Dictyostelium* cells. Gip1 regulates G-protein localization between the cytosol and plasma membrane upon receptor activation, which targets cytosolic G proteins to the membrane in a biased manner along chemoattractant gradients at higher chemotactic ranges. These findings provide evidence for a wide-range sensing mechanism in which Gip1-dependent G-protein

Significance

Eukaryotic chemotactic cells can recognize chemical gradients over a wide range of concentrations. This ability is physiologically important for numerous biological processes; however, its underlying mechanism is unknown. Here we report that the dynamic range of chemotaxis is extended to higher concentrations by gradient sensing achieved via regulation of trimeric G-protein shuttling between the cytosol and plasma membrane. G protein-interacting protein 1 (Gip1) regulates this intracellular G-protein translocation, which redistributes cytosolic G proteins to the plasma membrane along chemical gradients at high chemoattractant concentrations. This dynamic spatiotemporal regulation of trimeric G protein yields proper processing of receptor-mediated signaling.

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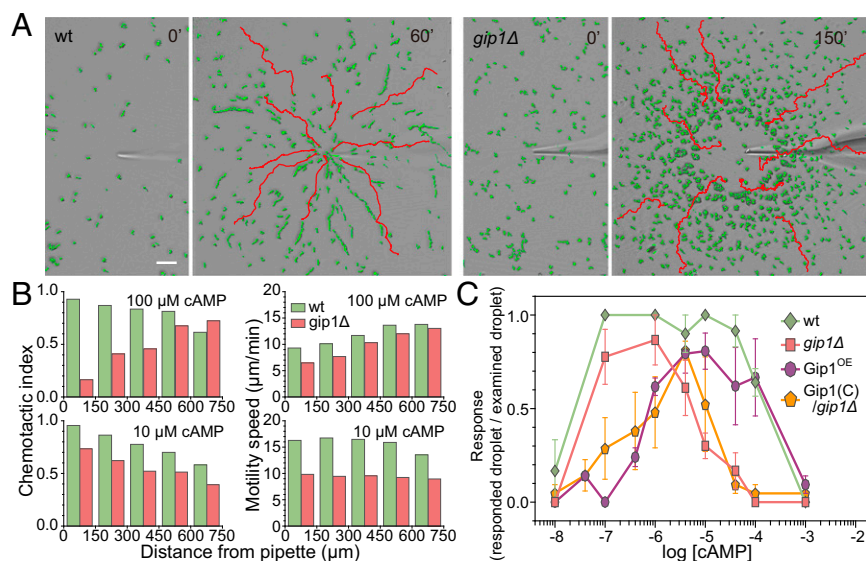


Fig. 2. Chemotactic dynamic range is extended to higher concentrations via Gip1. (A) Chemical gradients were applied to WT and *gip1Δ* cells from a micropipette tip filled with 100 μ M cAMP. Representative images of cell trajectories (red lines) are shown before (0') and 60 or 150 min after (60' and 150', respectively) the start of the assay. Cells are highlighted in green. (Scale bar, 50 μ m.) (B) CI and motility speed were calculated from the assay in A following application of 10 or 100 μ M cAMP at different distances from the micropipette tip. Each bar represents the average of at least 10 cells. (C) The chemotactic response to different cAMP concentrations was evaluated by the small population assay (Fig. S2B). Data represent the mean $\pm$ SEM of three experiments.

Uniform Receptor Stimulation Induced Normal Signaling Responses in *gip1Δ* Cells. To identify the defects in the chemotactic signaling of *gip1Δ* cells, we followed the chemoattractant-triggered signaling reactions (3, 4). Cells were uniformly stimulated with cAMP, and the phosphorylation of CAR1, G α 2, PKBR1, PKBA, and Erk2 as well as fluorescence resonance energy transfer (FRET) changes between G-protein subunits, PIP₃ production at the plasma membrane, and actin polymerization were evaluated (Fig. S3). As described in detail in *SI Text*, no obvious differences in these signaling reactions upon uniform cAMP stimulations were observed in *gip1Δ* or WT cells, showing that the uniform increase of cAMP from the basal level can induce normal activation of chemotactic signaling pathways in *gip1Δ* cells. Thus, cells can detect the initial increase of cAMP concentrations in a Gip1-independent manner.

Gip1 Sequesters Trimeric G Protein in Cytosolic Pools. We next investigated the subcellular localization of G proteins, given that they are modified upon cAMP stimulation (17). G α 2 and G γ conjugated with the fluorescent dye TMR (G α 2-TMR and TMR-G γ , respectively) via a Halo tag were used for localization analysis. G α 2-TMR and TMR-G γ were present in both the cytosol and plasma membrane of WT cells, whereas the cytosolic fractions of both proteins were significantly decreased in *gip1Δ* cells (Fig. 3A and Fig. S4A). Gip1^{OE} cells exhibited an increase in the cytosolic fraction of G proteins, which was also observed by expression of the Gip1 C terminus but not of the Gip1 N terminus. These observations were confirmed for endogenous G α 2 and G β by biochemical fractionation into cytosolic and membrane components (Fig. 3B). The cytosolic amounts of G α 2 and G β were significantly reduced to about 20% and 40%, respectively, in *gip1Δ* compared with WT cells. Overexpression of the Gip1 and Gip1 C terminus compared with vector control increased cytosolic G α 2 by 2.3-fold and 1.6-fold, respectively, and cytosolic G β by 3.8-fold and 2.0-fold. The lesser effect of Gip1 C terminus compared with full-length Gip1 is attributed to a difference in the expression level of these proteins (Fig. S4B). Thus, the cytosolic localization of G proteins requires Gip1, likely by complex formation via its C terminus in the cytosol (Fig. 1C). Consistently, the full-length and N and C termini of Gip1 were localized mostly in the cytosol (Fig. S4C). The biochemical fractionation assay confirmed that endogenous Gip1 as well as ectopically expressed full-length and the N and C termini of Gip1 were predominantly found in the supernatant fraction (Fig. S4C and D). Thus, Gip1 is responsible for maintaining the cytosolic pools of G proteins in the absence of chemoattractants.

Gip1 Regulates G-Protein Shuttling upon Receptor Stimulation. Because chemotaxis-competent cells redirect G proteins from the cytosol to the plasma membrane upon cAMP stimulation (17), we investigated whether Gip1 is involved in this process. In WT cells, G α 2 was clearly recruited to the membrane upon cAMP addition, which was assessed by the decreased levels of cytosolic G α 2-TMR (Fig. 4A and B, and Movie S6). Gip1^{OE} cells showed similar cAMP-induced G α 2 translocation, with a half-maximum at around 10 nM cAMP (Fig. 4A–C). However, the same was not true of *gip1Δ* cells even though there was less G α 2-TMR protein in the cytosol under resting conditions (Fig. 4A–C). Instead, the amounts of G α 2 on the membrane decreased slightly with the cAMP concentration in *gip1Δ* cells (Fig. 4B and C). Biochemical fractionation analysis of G α 2 localization confirmed these observations (Fig. 4D). Additionally, unlike G protein, Gip1 remained in the cytosol upon cAMP addition (Fig. S4E and F). Together with the detailed characterization described in *SI Text*, we concluded that cAMP-induced G α 2 translocation depends on Gip1 but not on Ras activation, PIP₃ production, or actin polymerization (Fig. S5).

Next we sought to identify the region of Gip1 required for G-protein translocation. Full-length Gip1 expression in *gip1Δ* cells rescued G α 2-TMR translocation upon cAMP stimulation (Fig. 4A and B and Movie S6). The Gip1 N terminus did not revert G α 2 to the resting-state cytosolic localization, nor did it rescue cAMP-induced G α 2 translocation (Fig. 4A and B). In contrast, the C terminus caused cytosolic retention of G α 2, an effect that was unaltered by cAMP stimulation (Fig. 4A–C). The C terminus is therefore sufficient for tethering G proteins in the cytosol but insufficient for inducing G-protein translocation to the membrane upon cAMP stimulation, suggesting regulatory roles of the N-terminal PH domain in G-protein translocation. Sequestration of cytosolic G protein via Gip1 C terminus is inhibitory for chemotaxis because the proteins available for receptor-mediated signaling are depleted from the membrane. Upon receptor stimulations at the higher concentration ranges, G protein translocates to the membrane, probably by releasing from Gip1, thus serving as a possible receptor-mediated signal (see *Discussion* for details).

***gip1Δ* Cells Fail to Process Gradient Signals at Higher Ranges.** As shown in Fig. 2, *gip1Δ* cells exhibited decreased chemotaxis without severe motility defects, suggesting that Gip1-dependent G-protein translocation is involved in gradient sensing. To estimate the extent of gradient sensing, PIP₃ production was evaluated by using the specific probe PH_{AKT}-GFP (21, 22). When 10 μ M cAMP was applied as a gradient by a micropipette to WT cells, a crescent-like

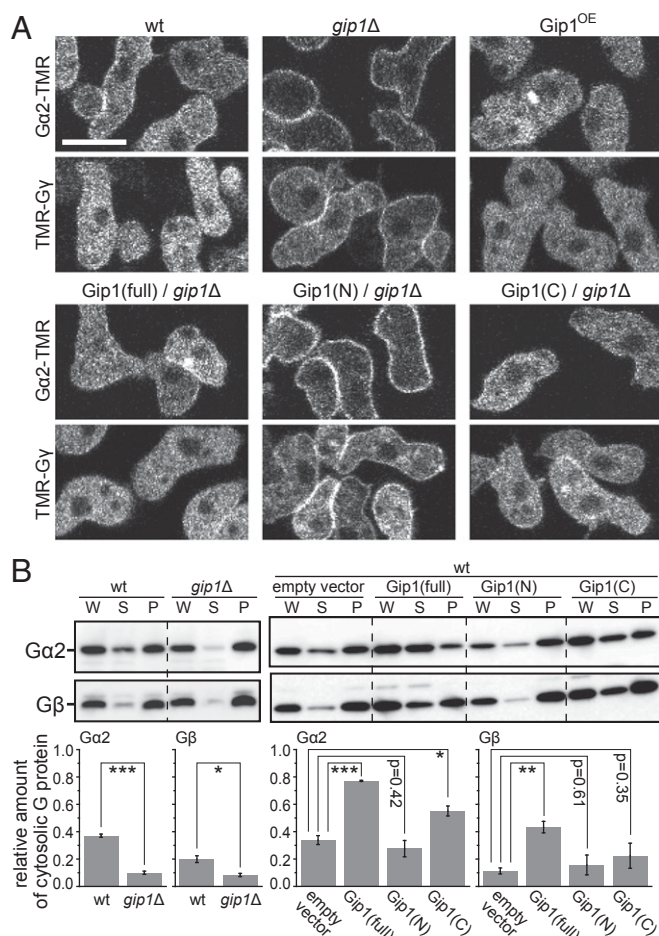


Fig. 3. Gip1 regulates shuttling of trimeric G protein between the cytosol and the plasma membrane. (A) Intracellular distribution of Gα2-TMR and TMR-Gγ, respectively. Gip1(full), Gip1(N), and Gip1(C) represent the coexpression of the full-length (full), N terminus, and C terminus of Gip1, respectively. (Scale bar, 10 μm.) (B) WCEs (W) prepared from the indicated cell lines were separated into supernatant (S) and pellet (P) fractions. Endogenous Gα2 and Gβ were detected by immunoblotting (Top). The fraction of the supernatant was calculated by band intensities (Bottom). **P* < 0.05, ***P* < 0.01, ****P* < 0.005, *t* test.

accumulation of PH_{AKT}-GFP was observed in areas close to higher cAMP concentrations (Fig. 5A, Fig. S64, Upper Left, and Movie S7), whereas *gip1Δ* cells did not exhibit the same bias in PH_{AKT}-GFP localization under cAMP gradients with high background concentrations (Fig. 5A, Fig. S64, Upper Right, and Movie S8). In lower cAMP concentrations supplied by a micropipette containing 0.1 μM cAMP, crescent-like distributions of PH_{AKT}-GFP near the plasma membrane were observed in WT cells and also in *gip1Δ* cells but with less efficiency (Fig. 5A and Fig. S64, Lower panels). To further confirm the defects in gradient sensing of *gip1Δ* cells, we used Ras binding domain (RBD)-GFP to detect activated Ras, which is another reporter for the cAMP signaling pathway (4). RBD-GFP behaved similarly to PH_{AKT}-GFP in WT and *gip1Δ* cells (Fig. S6B and C). These results indicate that Gip1 is required for gradient sensing at higher chemoattractant concentrations.

We next examined the spatiotemporal dynamics in PH_{AKT}-GFP localization upon gradient stimulation. Gradient stimulation initially caused a uniform increase in PH_{AKT}-GFP signals along the entire contour of WT cells within 10 s, and this effect was followed by an abrupt signal reduction on the membrane (Fig. 5B, Upper panels), which is due to the excitable property of PIP₃ production (10, 23, 24). These initial responses were similar in WT and

gip1Δ cells (Fig. 5B), which is consistent with the kinetics of PIP₃ production upon uniform stimuli (Fig. S3C). Differences between WT and *gip1Δ* cells were observed after the initial responses, the so-called second response. The restricted accumulation of PH_{AKT}-GFP was induced at the high side of the gradient in WT cells showing gradient sensing but not in *gip1Δ* cells (Fig. 5B). The diminished signals were never retained in *gip1Δ* cells. These data indicate that *gip1Δ* cells can detect information about the initial increase in cAMP but not gradient information with sufficient accuracy at higher chemoattractant ranges.

We further assessed Gip1-dependent Gα2 translocation under cAMP gradients by using a theta glass pipette with two separate chambers containing 0 and 10 nM cAMP solutions, respectively (Fig. 5C, Top Left). In WT cells, Gα2-TMR accumulated in the area stimulated with a higher cAMP concentration (Fig. 5C, arrowheads). The fluorescence intensities of Gα2-TMR in the stimulated region were 1.4-fold higher than in the unstimulated region. In contrast, the uniform distribution of the fluorescence signals along the entire membrane was not changed in *gip1Δ* cells upon gradient stimulation (Fig. 5C). That is, receptor stimulation induced Gα2 accumulation at the stimulated region of the membrane in a Gip1-dependent manner. Thus, Gip1-dependent G-protein translocation can underlie the availability and biased redistribution of G protein at the membrane under chemical gradients at the higher range. Moreover, the full availability of G protein to the membrane is insufficient for efficient gradient

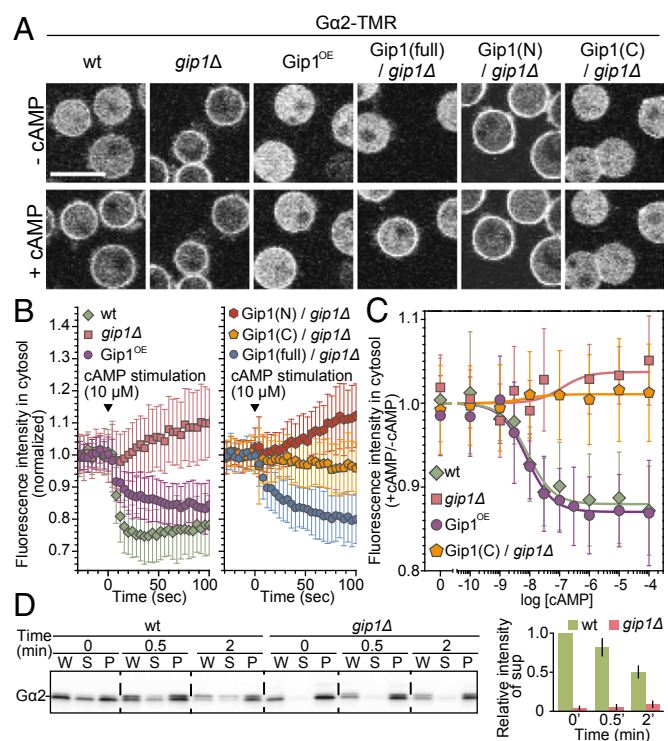


Fig. 4. Gip1 modulates spatial regulation of trimeric G protein in response to chemoattractants. (A) Fluorescent images of Gα2-TMR. Cells were treated with cAMP in the presence of 5 μM latrunculin A. (Scale bar, 10 μm.) (B) Cells expressing Gα2-TMR were stimulated with cAMP, and fluorescence intensity in the cytosol (mean ± SD) was quantified at 4-s intervals. (C) Translocation of Gα2-TMR to the plasma membrane was assessed quantitatively by the decrease in cytosolic fluorescence intensity. Gα2-TMR protein was expressed in WT and *gip1Δ* cells harboring Gip1-GFP (full-length and C terminus). A dose-dependent curve was plotted (mean ± SD). (D) Cells were fractionated before and after stimulation with 1 μM cAMP for 0.5 and 2 min (Left). The signal intensity of the supernatant fraction relative to that of WT cells at 0 min was quantified (Right).

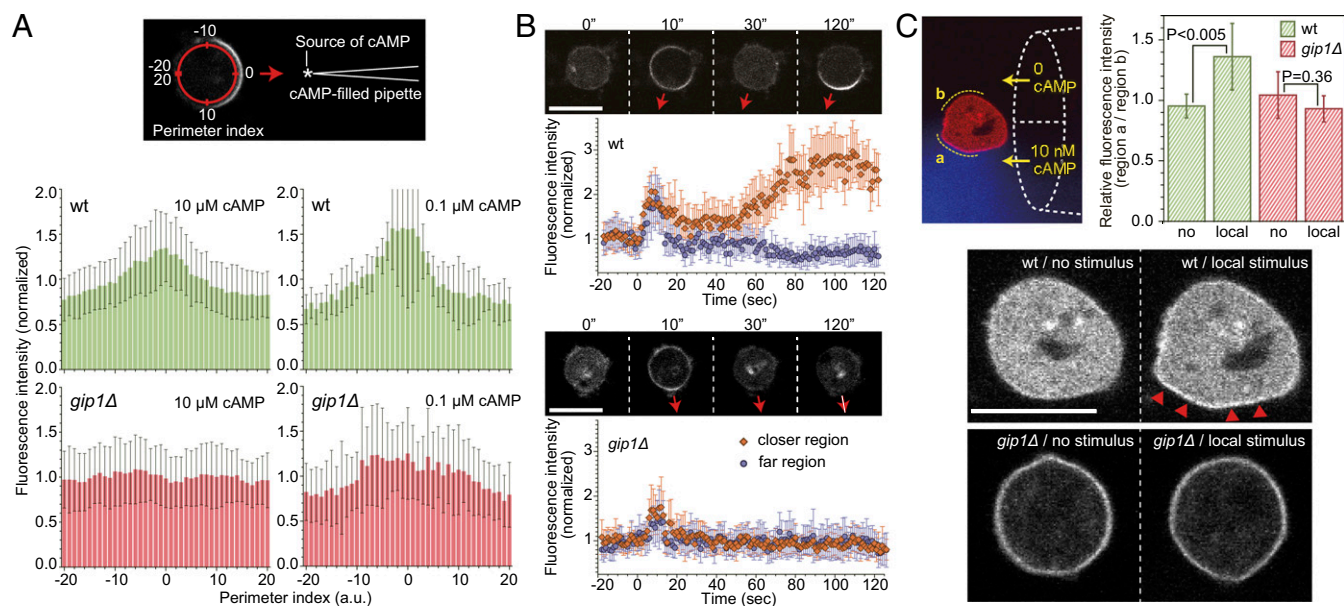


Fig. 5. GIP1 is involved in the signal transduction of chemical gradients. (A) Gradient sensing monitored by a PIP₃ probe, PH_{AKT}-GFP. Cells were treated with 5 μ M latrunculin A and stimulated with a micropipette tip containing 10 or 0.1 μ M cAMP. The cell perimeter was divided into 40 points (Top); relative intensities (mean $\pm$ SD) of PH_{AKT}-GFP in WT and *gip1* Δ cells are shown as histograms in green and red, respectively. (B) Representative temporal changes of PH_{AKT}-GFP localization in WT (Top) and *gip1* Δ (Bottom) cells when stimulated by a micropipette containing 10 μ M cAMP. Red arrows show the direction of the gradient source. Fluorescence intensities at the near (orange diamond) or far (blue circle) side of a cell from the micropipette were plotted over time as the mean $\pm$ SD. (Scale bar, 10 μ m.) (C) G α 2 localization under steep cAMP gradients. Cells expressing G α 2-TMR were stimulated by a theta micropipette with two separate chambers containing either 0 or 10 nM cAMP (Top Left), and G α 2-TMR localization was visualized (Bottom). The left and right images show pre- and poststimulation, respectively. Ratios of the fluorescence intensities in the regions a and b were measured (Top Right, mean $\pm$ SD). (Scale bar, 10 μ m.)

sensing at the higher range, as seen by *gip1* Δ cells lacking the translocation-mediated redistribution.

Such defects in gradient sensing by *gip1* Δ cells were also observed in moving cells. F-actin, as monitored by LimE Δ coil-RFP (25), was selectively localized at the higher cAMP gradient in WT cells, but the region of this signal was weakly restricted in *gip1* Δ cells (Fig. S6D). Similarly, RBD-GFP and PH_{AKT}-GFP were localized much more broadly in *gip1* Δ than in WT cells (Fig. S6D). These data suggest that GIP1 confines pseudopod formation along chemical gradients through the regulation of G-protein translocation for efficient chemotaxis.

Discussion

Identification of GIP1 as a Novel Regulator of Trimeric G Protein.

Using TAP of G β , we identified four interacting proteins with high specificity: G α , G γ , ElmoE, and GIP1 (Fig. 1A). ElmoE associates with G β and serves as a guanine nucleotide exchange factor for Rac to control F-actin in pseudopods (18). Because the interaction between G proteins and ElmoE was unaffected in *gip1* Δ cells (Fig. S1D), GIP1 and ElmoE function independently. We found GIP1 can bind to various G α subtypes, such as G α 2, G α 4, G α 5, and G α 9 (Fig. S1D and E), although phenotypic defects of *gip1* Δ cells were obvious only in G α 2-dependent chemotaxis in response to cAMP during early development. As discussed below, our analyses suggest GIP1 functions as a regulator of G-protein localization between the cytosol and the plasma membrane.

GIP1-Dependent G-Protein Translocation for Wide-Range Chemotaxis.

Previous studies have revealed that G proteins undergo translocation to the membrane upon receptor stimulation (17). We found that GIP1 regulates G-protein translocation by its sequestering function (Figs. 3 and 4) and is required for chemotaxis at higher ranges (Figs. 2 and 5). Furthermore, we found that receptor stimulation induced a biased redistribution of G protein under chemical gradients on the membrane in a GIP1-dependent manner (Fig. 5).

Based on these observations, we propose a model in which the role of GIP1 in G-protein translocation for wide-range chemotaxis can be explained by the following three steps (Fig. S7). First, GIP1 sequesters the cytosolic pools of G proteins through binding at the C terminus, where cytosolic G proteins are not available for chemotactic signaling on the membrane (Figs. 1 and 3). Second, receptor stimulation facilitates G-protein translocation from the cytosol to the membrane depending on the N-terminal PH domain of GIP1 (Fig. 4). Third, GIP1-dependent translocation supplies a signal transducer for chemoattractant receptors at higher ranges and causes a biased redistribution of G proteins on the membrane along chemical gradients (Fig. 5). This biased redistribution contributes to proper gradient sensing at higher concentration ranges.

Chemotaxis ranges observed in the cells with the knockout, overproduction, and truncation of GIP1 can be explained by G-protein dynamics in G-protein subcellular localizations—that is, sequestering in the cytosol, translocation to the membrane, and biased redistribution on the membrane. Because G protein on the membrane is responsible for chemotactic signaling, its cytosolic sequestering by GIP1 inhibits chemotaxis. G-protein translocation upon receptor stimulation can eliminate the inhibitory effects of GIP1, resulting in fully available G protein on the membrane. GIP1^{OE} cells have impaired chemotaxis at the lower but not higher ranges because of excessive sequestration but functional translocation of cytosolic G protein. On the other hand, C-terminal GIP1^{OE} cells have impaired chemotaxis at the lower and higher ranges because C-terminal GIP1 loses the translocation activity of G protein but still maintains sequestration activity. In the case of *gip1* Δ cells, G protein is fully available over a wide range but has an impaired biased redistribution of G protein, leading to chemotactic defects at the higher ranges. Thus, it is likely that the proper regulation of G-protein availability by GIP1 is a prerequisite for gradient sensing over a wide range. Recently, it was reported that the Ric8 homolog in *Dictyostelium* cells contributes to chemotaxis by amplifying G-protein activity at

lower cAMP concentrations (26). Thus, the wide sensitivity to chemical gradients could be achieved by multiple mechanisms, including receptor phosphorylation, Ric8, and Gip1.

Gip1-Dependent Gradient Sensing. We show evidence that Gip1 is involved in gradient sensing at higher chemotactic ranges (Fig. 5). Under uniform chemoattractant concentrations, *gip1Δ* cells exhibited receptor-induced reactions almost normally (Fig. S3). Upon gradient stimulations, *gip1Δ* cells initially distributed PIP₃ along the entire plasma membrane (Fig. 5B). These results indicate that a sudden increase of receptor stimulation can be transduced by G protein without Gip1 leading to the excitation of PIP₃, suggesting that STEN works normally without Gip1. This excitable reaction was subsequently followed by the biased accumulation of PIP₃ along chemical gradients in WT cells, whereas this event was not observed in *gip1Δ* cells (Fig. 5A and B). During the biased PIP₃ accumulation, cells obtained gradient signals by adapting to the average concentration of the chemoattractant across the cell body, which can be explained by a local excitation and global inhibition (LEGI) model combined with STEN (LEGI-STEN model) (10). LEGI works as an adaptive network that detects gradient signals of various concentrations by combining the excitor and the inhibitor, which reflect the local and the average concentration of the chemical gradient, respectively. STEN works as an amplifier of the LEGI output to generate a localized signal in an all-or-none fashion. According to the LEGI-STEN model, *gip1Δ* cells likely have defective LEGI but functional STEN, because the initial PIP₃ excitation was normal but the biased PIP₃ accumulation was not. Our observations also suggest that the Gip1-dependent translocation and redistribution of G protein on the membrane can contribute to a gradual localization of excitor in chemotactic signaling at higher chemotactic ranges, which expands the dynamic range for chemotaxis.

Materials and Methods

Chemotaxis Assay. Before the assay, cells were treated to proceed with their development as previously described (20). For the small population assay, ~3,000 developed cells in a 1-μL droplet on the nonnutrient agar plate was placed next to a drop of cAMP solution. The distance between the centers of the two droplets was 2 mm. After 30 min, a responded droplet was assessed by cell accumulation at the edge near a cAMP droplet. For the micropipette assay, 2×10^5 developed cells were seeded on a glass-bottom dish, and cAMP gradients were applied by a Femtotip microcapillary (Eppendorf) containing 10 or 100 μM cAMP and ATTO 532 (AD 532-21; ATTO-TEC GmbH) with a constant pressure of 10 hPa.

Gradient Sensing Assay by PH_{AKT}-GFP Measurements. Cells expressing PH_{AKT}-GFP were subjected to cAMP gradients by a Femtotip microcapillary containing 100 nM or 10 μM cAMP in the presence of 5 μM latrunculin A. Fluorescence images of PH_{AKT}-GFP were acquired by confocal microscopy at 1- or 10-s intervals. Fluorescence intensities were measured along the plasma membrane of each cell and then normalized by the average intensity of the perimeter.

Trimeric G-Protein Translocation. Starved cells expressing Gα2-TMR and TMR-γ were treated with 5 μM latrunculin A. Cells were stimulated uniformly by adding 200 μL of various concentrations of cAMP solution to 10 μL of a cell suspension droplet in a glass-bottom dish. Fluorescence images were acquired by confocal microscopy at 4-s intervals, and the intensities on the plasma membrane and the cytosol were measured.

Cell Preparations and Other Methods. *Dictyostelium discoideum* cells were used for all experiments. Full methods and any associated references are described in *SI Materials and Methods*.

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